Mad Scientist 1

Four simulation studies for COP greater than one

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Abstract

Four open problems invite a computational search for complete-cycle COP greater than one. A falling-weight lever, switched transformer, contact chain and coupled-coil release connect striking observations to explicit dynamical models. Each proposed study follows preparation, useful delivery and reset, integrating receiver work and every inward power flow. Exact examples supply reference cases; contact and switching laws define the territory to explore. The aim is to locate reproducible model behavior, resolve its energy transfers and identify the assumptions that make an above-one result possible.

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Local draft; the GitHub destination below is planned, not published.
Latest on GitHub: https://github.com/hobnilre/physics-mad-scientist-1

The top four

I want a model that raises a load or charges a receiver, restores its machine state, and makes every transfer inspectable. The ambition is a coefficient of performance (COP) greater than one: useful receiver work divided by **all gross inward energy** over the complete history. Give the output a load, give each input an integral, and let the trajectories reveal what happened.

This list alternates the two leading mechanical and two leading electrical questions, preserving their source order within each domain. Begin with the lever and transformer; their modeling and accounting tools carry into the chain and coil studies.

1. **The falling-weight lever.** Can contact timing and compliance produce receiver work exceeding the model's lifting, release and reset input? [OP-CT-HW-01]
2. **The switched transformer.** Can a finite tap-commutation model reach COP above one when source, driver and restoration work are included? [OP-TRF-02]
3. **The contact chain.** Can mass order, gaps and branching produce loaded COP above one through a complete launch-and-reset cycle? [OP-LR54-03]
4. **The coupled-coil release.** Can a resolved release path deliver more useful work than energization, switching and reset supply? [OP-LR48-03]

The clues are a fast launch, different voltages across different terminal pairs, and growing sums of momentum or flux-linkage magnitudes. Each gives a physical idea to explore. A simulation can expose the full state history, change one law at a time, and show exactly which transfer changes the COP.

These are proposed simulation studies, with exact analytic reference cases. Derive dynamics from forces, circuit constraints and component laws; evaluate energy independently. Follow promising regions by resolving events and identifying the responsible law. The final page gives the shared method.

1 The falling-weight lever

A teeterboard turns the downward arrival of one or two people into the upward flight of another. Timing, leg compression and board flexibility change the launch. Replacing performers with guided masses gives a compact model in which contact timing can be explored against a complete-cycle COP target.

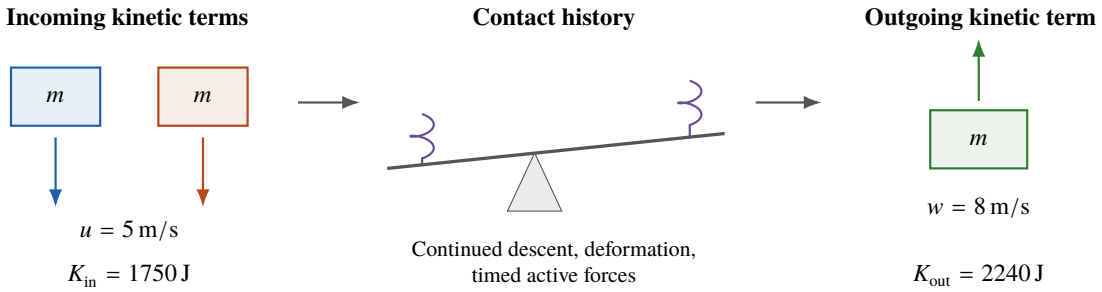


Figure 1. Illustrative endpoint comparison for three equal masses. The mechanism is schematic; descent and deformation continue during contact.

The observation. Let each mass be $m = 70\,\mathrm{kg}$. Two arrive downward at $u = 5\,\mathrm{m/s}$ and one receiver leaves upward at $w = 8\,\mathrm{m/s}$. The selected kinetic terms are

$$K_{\mathrm{in}} = mu^2 = 1750\,\mathrm{J}, \quad K_{\mathrm{out}} = \tfrac{1}{2}mw^2 = 2240\,\mathrm{J}, \quad K_{\mathrm{out}} - K_{\mathrm{in}} = 490\,\mathrm{J}. \tag{1}$$

Their ratio is $32/25$. The contact history connects that comparison to the COP question: descending masses continue losing gravitational potential energy, compliance stores and returns work, and active forces can act during contact.

What made the question open. An ideal fixed pivot supplies impulse at zero work because its point of application stays still. The board's rotation, recoil, catch engagement and separation time therefore matter independently of the outgoing speed. Models imposing the same takeoff speed can require different force and actuator-work histories. Those histories are the information to recover.

The passive model has its own scope, separate from the original human hardware question. Let the launch emerge from its equations. Change the arrival delay and contact compliance, then ask whether the useful delivery grows faster than all work needed to prepare and repeat it.

1.1 Simulation: let the launch emerge

Use driver heights y_i, lever angle ϕ, receiver height y_R and their velocities as states. A rigid lever has inertia I and its center of mass at a fixed pivot. At signed arm coordinate a_α, its contact height is $h_\alpha = a_\alpha \sin\phi$. Define compression $\delta_\alpha = h_\alpha - y_\alpha - d_\alpha$, with offset d_α. These definitions determine the contact motion rather than prescribing a launch.

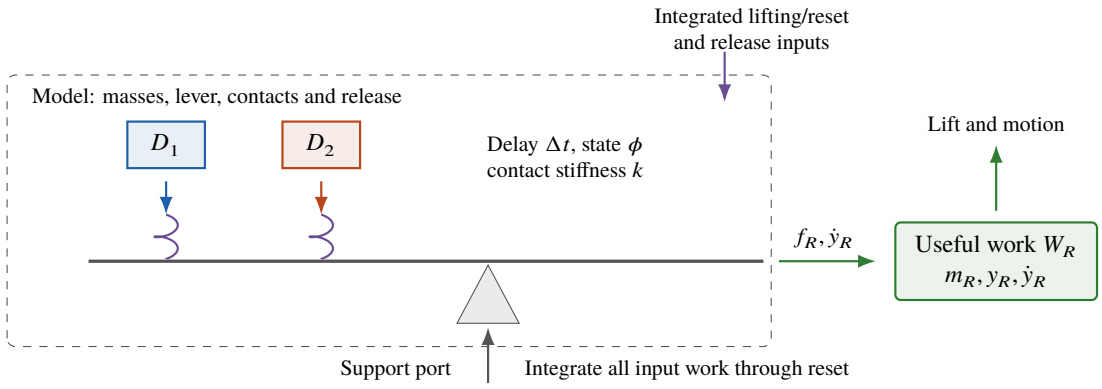


Figure 2. Lever simulation model, schematic. Evolve masses, rotation and contact states; integrate actuator inputs and work at the outside receiver.

Specify the contact law. Start with $f_\alpha = k_\alpha[\delta_\alpha]_+$, where $[x]_+ = \max(x, 0)$ and f_α acts upward on the contacting mass. Newton's equations give

$$m_i\ddot{y}_i = -m_i g + f_i + u_i, \quad m_R\ddot{y}_R = -m_R g + f_R, \quad I\ddot{\phi} = \tau_a - \sum_\alpha f_\alpha h'_\alpha(\phi). \tag{2}$$

Here u_i and τ_a are preparation/reset actuator efforts; the sum includes driver and receiver contacts. Prescribe finite actuator laws and integrate their work. A latch or release actuator needs its own law and port. First use elastic contacts, then declare a dissipative alternative and account for its heat.

Calculate the COP. Integrate the complete cycle from a ready state through lifting, release, relaxation and driven restoration:

$$W_R = \int_{t_0}^{t_1} f_R\dot{y}_R\,dt, \qquad \mathrm{COP}_L = \frac{W_R}{D_{\mathrm{act}} + D_{\mathrm{release}} + D_{\mathrm{other}}}. \tag{3}$$

Each D is gross inward port work. Form D_{act} from the separate powers $u_i\dot{y}_i$ and $\tau_a\dot{\phi}$, using the common rule on the final page. Include any added support or control supply in the appropriate input. Check receiver work against $m_R g\Delta y_R + \frac{1}{2}m_R\Delta(\dot{y}_R^2)$ in this ideal receiver model.

Evaluate the machine's driver kinetic/gravitational, lever rotational and contact-spring stores from its states. Include the receiver-contact spring inside the machine but the receiver mass outside. Reposition through actuator dynamics; retain every transfer through reset.

Explore and resolve. Compare one and two drivers, equal and unequal arrival times, contact stiffness and loaded receiver mass. Hold the prepared state fixed within each timing comparison while retaining its preparation cost. Locate every engagement and release event. Refine the timestep at fixed stiffness before investigating harder contacts as a separate physical limit. Report input, output, COP and state-return error together. Follow any persistent above-one region through an independent contact formulation and its explicit work account.

2 The switched transformer

A transformer tap offers several places to connect a receiver. The motivating analogy is a planet gear: rotation relative to the carrier differs from rotation relative to the ring. Electrically, the terminal pair determines the voltage. A timed return connection is therefore a concrete variable in the COP search.

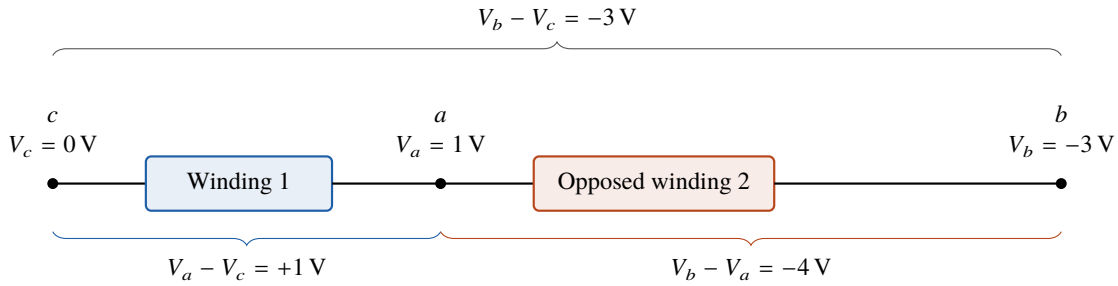


Figure 3. Same instantaneous ideal state, shown schematically. Opposed winding voltages add algebraically and give different receiver-terminal voltages.

The observation. Winding 1 joins a, c and winding 2 joins b, a. Along one electrical trajectory,

$$V_b - V_c = (V_b - V_a) + (V_a - V_c). \tag{4}$$

An opposed ideal ratio $n = -4$ gives $V_b - V_a = n(V_a - V_c)$. At $V_a - V_c = 1\,\mathrm{V}$, the two receiver voltages are $-4\,\mathrm{V}$ across $b - a$ and $-3\,\mathrm{V}$ across $b - c$. The idea is to use different connections during different parts of a cycle.

A lightly loading meter can observe both on essentially the same state. A loaded return connection changes the current history. A common potential-zero shift leaves voltage differences unchanged; commutation changes the circuit itself. A simulation must represent that change in its equations.

What made the question open. Fixed-connection descriptions leave the transition to be specified. Winding current, capacitor charge, clamp conduction and selector drive all participate. Existing finite models cover particular transitions; physical switch and gap laws remain questions for the model. One bounded complete-history comparison required more supply after switching at matched receiver service. That result supplies a specific comparison to extend.

The opportunity is to vary the actual transition law and timing, follow every supply and receiver transfer, and determine which choices raise complete-cycle COP. A voltage comparison starts the question; a resolved loaded history answers it for the chosen model.

2.1 Simulation: evolve the finite switching network

Build the tapped circuit with winding resistance, a reciprocal inductance matrix, winding/node capacitances and finite selector and clamp paths. Use winding currents, capacitor voltages and driver states as dynamic variables. The return moves between a and c through a declared gate waveform, with an explicit RC driver and its power supply included.

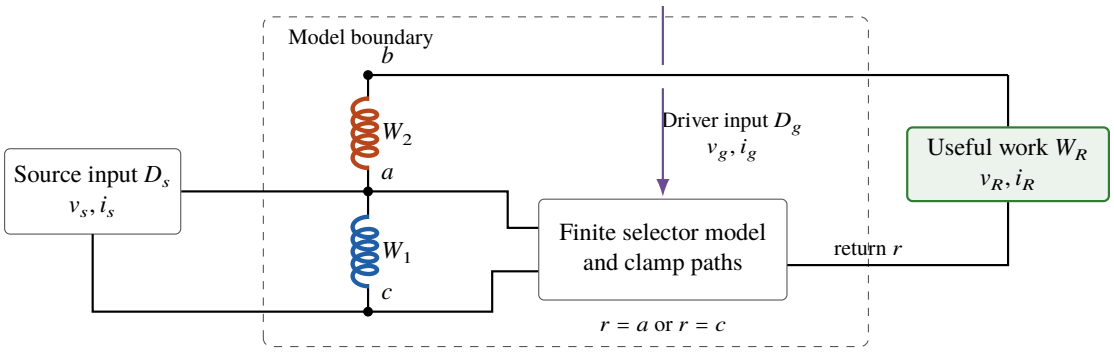


Figure 4. Transformer simulation topology. The finite selector and driver are modeled elements; source and receiver powers come from the resulting trajectories.

Assemble the equations. With one node grounded, a useful network form is

$$\mathbf{L}\dot{\mathbf{i}} + \mathbf{R}\mathbf{i} = \mathbf{A}^T\mathbf{v}, \qquad \mathbf{C}_n\dot{\mathbf{v}} + \mathbf{A}\mathbf{i} + \mathbf{G}_\sigma(\mathbf{v})\mathbf{v} = \mathbf{j}_{\text{ext}}. \tag{5}$$

$\mathbf{A}$ has $+1$ at each winding's current-entry node and -1 at its exit. $\mathbf{C}_n$ is the nodal capacitance matrix; $\mathbf{G}_\sigma$ represents resistive, switch and clamp paths in state σ. External sources and the receiver supply the signed injections $\mathbf{j}_{\text{ext}}$. Keep finite on/off conductances and transition duration explicit. Choose winding signs consistently, with $L_1, L_2 > 0$ and $M^2 < L_1L_2$ for the two-winding model.

Compare useful service. Simulate fixed return at a, fixed return at c, and finite commutation. Vary switching phase and transition duration independently of numerical step size. Use a declared source-amplitude adjustment to match receiver work and delivery time at a fixed load; retain cases that cannot match. Compare make-before-break and break-before-make paths as separate topologies.

Integrate the result. With j_R entering the outside receiver at b,

$$W_R = \int_{t_0}^{t_1} (V_b - V_r) j_R \, dt, \qquad \text{COP}_T = \frac{W_R}{D_s + D_g + D_{\text{other}}}. \tag{6}$$

D_s is gross source input and D_g sums independent driver inputs. Integrate from before energization through relaxation and simulated restoration. Check winding, capacitor and driver states; carry dissipated energy as heat or an explicit outward transfer. A constant-capacitance receiver gives the independent ideal check $W_R = C_R(V_{R,f}^2 - V_{R,i}^2)/2$.

Resolve gate edges and clamp onset with a stiff-system solver, then compare an independent method and finer power quadrature. Track the energy residual without forcing it to vanish. Report absolute COP beside improvement over the fixed-tap controls. A persistent above-one result identifies a particular netlist, transition law and parameter region for deeper examination.

3 The contact chain

Unequal masses and compliant contacts can pass a disturbance along a chain. Some masses reverse while others advance; a light terminal mass can become fast. The COP ambition is to turn those exchanges into useful receiver work exceeding the complete launch and restoration input.

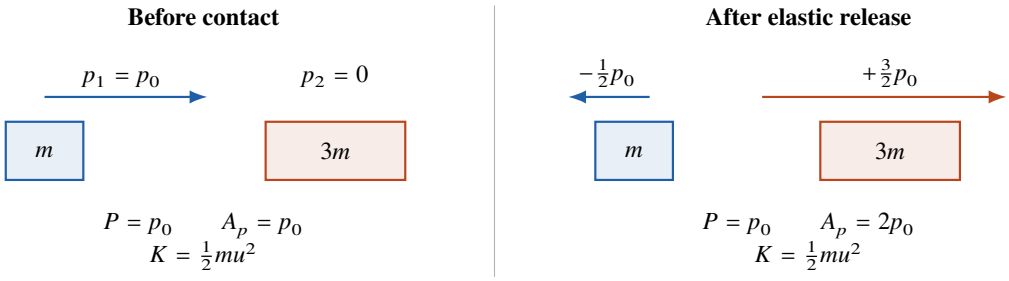


Figure 5. Exact two-mass elastic-contact illustration, with $p_0 = mu > 0$. Momentum arrows show how opposite motion doubles the magnitude sum while the signed total stays fixed.

The observation. A mass m arrives at $u > 0$; a mass $3m$ is initially still. For a lossless Hooke contact of stiffness k, compression δ obeys $m\dot{v}_1 = -k\delta$, $3m\dot{v}_2 = k\delta$ and $\dot{\delta} = v_1 - v_2$. Starting at $\delta = 0$ gives

$$v_1 = \frac{u}{4}(1 + 3\cos\theta), \quad v_2 = \frac{u}{4}(1 - \cos\theta), \quad \theta = t\sqrt{\frac{4k}{3m}}. \tag{7}$$

At $\theta = \pi$, compression returns to zero. The separating momenta are $-p_0/2$ and $3p_0/2$: their signed sum is p_0, their magnitude sum $2p_0$. Endpoint kinetic energy is $\frac{1}{2}mu^2$ in both cases.
For a free chain, $A_p = \sum_j |p_j| = P + 2\sum_{p_j<0} |p_j|$, with $P = \sum_j p_j$. Internal forces preserve P while opposing motion can increase A_p. Receiver work and the independently evaluated stores connect that motion to the COP question.

What made the question open. A permanent spring can pull as well as push; a contact releases. Gaps change arrival times, branches change the contact network, and restitution, plasticity and fixture motion change the laws. The original free, zero-gap compliant descriptions leave those extensions open. Increasing stiffness also raises a separate rigid-impact-limit question.

Choose mass order and spacing to make the terminal contact deliver at a useful moment. Its receiver reacts back on the chain. A simulation can follow every mass and contact through that feedback while calculating full-cycle COP.

3.1 Simulation: follow the changing contact network

Begin with three translating masses and unilateral spring contacts. Declare mass ratios, gaps and the receiver load before exploring timing; a ratio such as $1 : 3 : 1/2$ is an illustrative starting configuration. Prepare incidence through a finite launcher force, then evolve the contacts and receiver together.

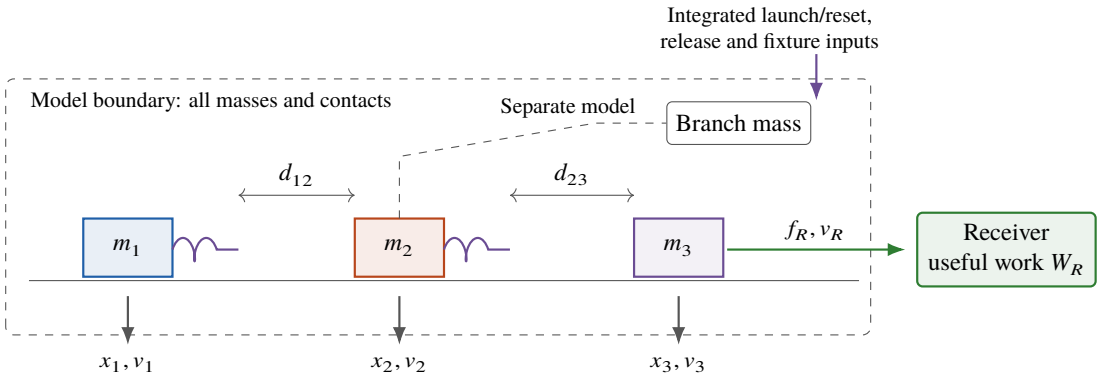


Figure 6. Contact-network simulation, schematic. Integrate all mass states, contact events, launcher/reset work and useful receiver delivery. The branch is a separate model configuration.

Evolve the states. For the line, let $\mathbf{B}$ have -1 at an edge's left mass and $+1$ at its right mass, and let $\mathbf{M} = \mathrm{diag}(m_j)$. Then

$$\dot{\mathbf{x}} = \mathbf{v}, \qquad \mathbf{M}\dot{\mathbf{v}} = \mathbf{B}\mathbf{f} + \mathbf{u} - \mathbf{e}_N f_R, \qquad \dot{\delta} = -\mathbf{B}^T\mathbf{v}. \tag{8}$$

$\mathbf{u}$ contains external actuator forces; $\mathbf{e}_N$ selects the terminal mass. Offset each compression by its initial gap. Set $f_e = k_e[\delta_e]_+$, with $[x]_+ = \max(x, 0)$, as the first contact law. For a dissipative variant, add $c_e[\dot{\delta}_e]_+$ only while $\delta_e > 0$, where $c_e \geq 0$. Its dissipated power is $c_e[\dot{\delta}_e]_+^2$ during contact.
Couple the terminal mass to an outside mass-spring receiver through an explicit receiver-contact law. Include that contact's spring inside the machine. For a branch, replace the line incidence relation by the Jacobian of the declared contact geometry and track the extra mass. Compare laws and geometries as distinct models.

Calculate useful delivery. At the receiver interface,

$$W_R = \int_{t_0}^{t_1} f_R v_R \, dt, \qquad \mathrm{COP}_C = \frac{W_R}{D_{\text{launch}} + D_{\text{release}} + D_{\text{fixture}} + D_{\text{other}}}. \tag{9}$$

Integrate preparation and repositioning forces through the whole history. Evaluate $\sum_j m_j v_j^2/2 + \sum_e k_e[\delta_e]_+^2/2$ independently, adding receiver-contact and any fixture stores. Track heat for dissipative contacts. Check an ideal receiver against its kinetic and spring-energy change.
Search with controls. Compare zero and nonzero gaps, reverse the mass order, add one declared branch, and change receiver loading. First reproduce the exact two-mass contact above. Locate engagement and force-release events, retain event-side states and refine integration at fixed contact parameters. Increase stiffness only in a separate sequence. Drive the simulated reset and verify all masses, springs and thermal states return within declared tolerances. Report COP, input, output and reset mismatch for every case; investigate an above-one region by isolating the contact law and event sequence that produced it.

4 The coupled-coil release

Two inductors exchanging through a capacitor provide an electrical counterpart to the chain. Their flux linkages can acquire opposite signs while their signed sum stays fixed. A growing magnitude sum suggests a next step: change the connection at a chosen moment and route the state into a receiver.

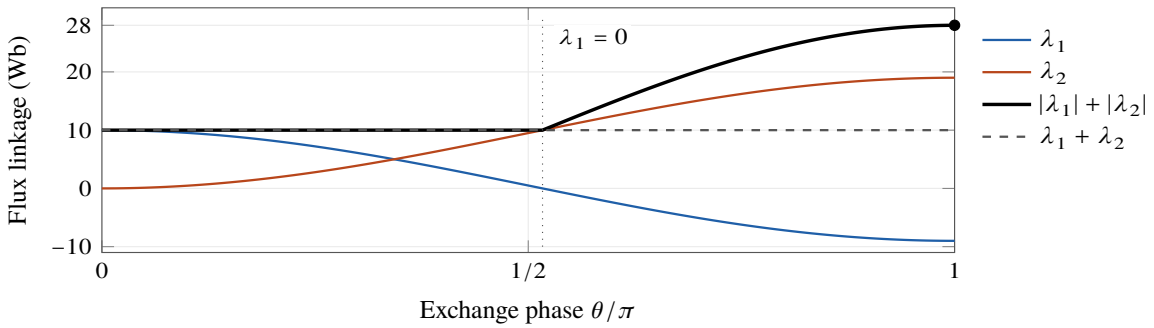


Figure 7. Exact connected-model curves for the stated zero-coupling illustration. Opposite linkage signs separate the growing magnitude sum from the fixed signed sum.

The observation. In the zero-mutual-coupling limit, take $L_1 = 1\,\mathrm{H}$, $L_2 = 19\,\mathrm{H}$, $C > 0$, initial currents $i_1 = 10\,\mathrm{A}$ and $i_2 = 0$, and zero capacitor voltage. For equal-and-opposite winding voltages,

$$i_1 = \frac{1 + 19\cos\theta}{2}\,\mathrm{A}, \quad i_2 = \frac{1 - \cos\theta}{2}\,\mathrm{A}, \quad \theta = \omega t, \quad \omega^2 = \frac{1/L_1 + 1/L_2}{C}. \tag{10}$$

The plotted linkages are $\lambda_j = L_j i_j$. At $\theta = \arccos(-1/19)$, the first crosses zero. By $\theta = \pi$, the pair has moved from $(10, 0)$ to $(-9, 19)$ Wb. Their signed sum stays 10 Wb while the magnitude sum reaches 28 Wb. The capacitor is again at zero voltage; magnetic storage is 50 J initially and $(81 + 19)/2 = 50\,\mathrm{J}$ finally.

What made the question open. Release changes the circuit that linked the winding voltages. At finite current, the switch gap, clamp and winding capacitance provide the next paths; arc behavior may add further state. The connected solution supplies a precise starting point for asking what those paths deliver.

Choose release time, coupling and winding orientation as design variables. The COP question compares useful delivery with all work needed to prepare, switch and restore the modeled circuit. The exact connected solution becomes a reference test for that larger computation.

4.1 Simulation: resolve the release and its return cycle

Use currents, capacitor voltages, gate states and any declared clamp/core states. Fix winding geometry within each cycle and represent both switches with finite conductances and finite transitions. Figure 8 supplies the model topology and the receiver boundary.

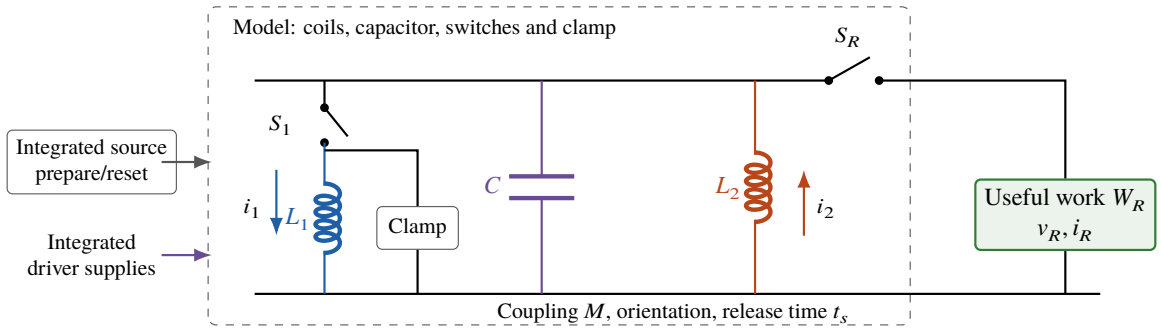


Figure 8. Coil-release simulation topology. Preserve state through finite switching; integrate preparation/reset, driver and receiver power from the same history.

Build the connected phase first. With winding resistances R_1, R_2, consistent opposed current references and signed mutual inductance M,

$$\begin{pmatrix} L_1 & M \\ M & L_2 \end{pmatrix} \begin{pmatrix} \dot{i}_1 \\ \dot{i}_2 \end{pmatrix} = \begin{pmatrix} V_C - R_1 i_1 \\ -V_C - R_2 i_2 \end{pmatrix}, \qquad C\dot{V}_C = i_2 - i_1. \tag{11}$$

Take $L_1, L_2 > 0$ and $M^2 < L_1 L_2$. Reproduce the analytic $M = R_1 = R_2 = 0$ reference before adding release. During release, solve the new node equations with the same winding constitutive laws; the two winding voltages need no longer be opposite. Retain currents and capacitor voltages through finite transitions.

Give every path a law. Start with winding resistance, capacitance and a specified clamp current-voltage law. Drive the gates through explicit source, resistance and capacitance models. Add a gap/arc state or nonlinear core only as a separately declared extension. Its state evolution and power exchange must be supplied. Sweep release phase and both orientations. An illustrative coupling set is $|M|/\sqrt{L_1 L_2} \in \{0, 1/2, 3/4\}$. Compare zero-current and finite-current release with equal prepared states.

Compute the COP and stores. Integrate receiver power and each gross input through energization, exchange, release, relaxation and modeled reset:

$$W_R = \int_{t_0}^{t_1} v_R i_R \, dt, \qquad \mathrm{COP}_{\mathrm{coil}} = \frac{W_R}{D_{\mathrm{source}} + D_{\mathrm{driver}} + D_{\mathrm{other}}}. \tag{12}$$

Compute the joint store $U_M = L_1 i_1^2/2 + M i_1 i_2 + L_2 i_2^2/2$ and capacitor stores from the states. For fixed inductances, $\dot{U}_M = i_1 \dot{\lambda}_1 + i_2 \dot{\lambda}_2$. A constant-capacitance receiver gives the independent ideal check $W_R = C_R(V_{R,f}^2 - V_{R,i}^2)/2$. Carry winding/clamp losses into thermal states or outward heat, and integrate the actual reset trajectory.

Refine event location, timestep and quadrature while holding component laws fixed. Then vary clamp and stray capacitance as physical model choices. Report COP and numerical sensitivity beside the restored-state error. A persistent above-one region invites a direct comparison of the models: which release path changed useful output, which input paid for it, and which law accounts for the remaining difference?

The same calculation in every model

Declare the machine boundary and all state variables before integrating. At each crossing port k, use inward-positive power P_k: vi electrically, $\mathbf{F} \cdot \mathbf{v}$ mechanically, or $\tau \cdot \omega$ rotationally. Effort and motion belong to the same station and frame. Count shared supplies once. Include any thermal or environmental input; keep the useful receiver outside.

For the complete interval $[t_0, t_1]$, calculate each port separately:

$$D_k = \int_{t_0}^{t_1} [P_k]_+ \, dt, \qquad X_k = \int_{t_0}^{t_1} [-P_k]_+ \, dt, \qquad W_k = D_k - X_k, \tag{13}$$

$$E_{\text{in}} = \sum_k D_k, \qquad E_{\text{out}} = \sum_k X_k, \qquad r_E = \Delta U - E_{\text{in}} + E_{\text{out}}. \tag{14}$$

Here $[x]_+ = \max(x, 0)$ and ΔU is evaluated independently from endpoint states. Internal contact and clamp transfers explain the history; they are counted as boundary transfers only when they cross the chosen boundary. Retained heat is a store, exported heat a transfer.

Let W_R be net useful work into the declared receiver after return flow. All receiver-to-machine work also contributes to gross input. Report source return separately. With a resolved positive denominator,

$$\text{COP} = \frac{W_R}{E_{\text{in}}}, \qquad \text{COP}_{\text{min}} = \frac{W_R - \varepsilon_R}{E_{\text{in}} + \varepsilon_{\text{in}}}. \tag{15}$$

The second expression is a lower bound only when nonnegative error bounds $\varepsilon_R, \varepsilon_{\text{in}}$ are justified simultaneously for the fixed model and $W_R - \varepsilon_R > 0$. Differences between refined runs are numerical error estimates, not automatically certified bounds. Declare an input floor before comparing ratios. Keep component-law uncertainty separate.

Simulate the complete history. Begin before preparation from a declared ready state. Evolve preparation, operation, contact/switching, relaxation and reset. Return through modeled forces and circuits, recording their work; compare all relevant endpoint states with stated tolerances. Bound any remaining depletion independently. Keep delivered work outside while accounting for any backflow.

Make the numerics answerable. Check exact reference cases first. Locate contact and switch events, split power integration there, and retain both event-side states. Refine timestep, solver tolerances and quadrature independently; compare another numerical method or formulation. Keep physical stiffness and switch duration fixed during refinement. Study their limiting values separately. Evaluate the energy residual afterward, without projecting states onto a desired balance. Explain any persistent residual within the declared laws and boundary.

Follow the promising region. Freeze parameter ranges, loads and comparison rules before the first sweep. Preserve failures, both residual signs and reset mismatches. Follow a stable above-one region with independent runs, broader component-law checks and a reconstructed transfer history. A verified bound above one establishes a result for that specified model; its physical realization becomes the next question. Save the model, parameters, event history and work integrals so another researcher can reproduce the finding and pursue it.